

Kindle Agent - Technical Requirements Document (TRD)

1. System Architecture

User Interface ↓ Agent Layer ↓ MCP Server ↓ Tool Layer ↓ External Utilities

2. Technology Stack

Backend

- Python 3.12+
- FastMCP SDK

Agent Framework

- OpenAI Agents SDK or LangGraph

OCR

- Tesseract OCR
- pytesseract

PDF Processing

- PyMuPDF
- pdfplumber

Conversion

- Calibre CLI
- ebook-convert

Storage

- Local filesystem (initial version)
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3. MCP Server Design

Server Name:

Kindle Agent MCP Server

Exposed Tools

analyze_pdf

Input:

- PDF path

Output:

- PDF type
- Page count
- Metadata

run_ocr

Input:

- PDF path

Output:

- Searchable PDF

extract_metadata

Input:

- PDF path

Output:

- Title
- Author
- Subject

convert_to_epub

Input:

- PDF path

Output:

- EPUB file

convert_to_azw3

Input:

- EPUB path

Output:

- AZW3 file
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4. Agent Workflow

Step 1: Analyze PDF

Step 2: Determine OCR necessity

Step 3: Run OCR if required

Step 4: Extract metadata

Step 5: Convert to EPUB

Step 6: Convert to AZW3

Step 7: Return result

5. Folder Structure

kindle-agent/

```
|—— server.py |—— tools/ | |—— analyze_pdf.py | |—— run_ocr.py | |—— metadata.py | |——  
convert.py |—— uploads/ |—— outputs/ |—— requirements.txt |—— README.md
```

6. Error Handling

Invalid PDF

Return:

- Validation error

OCR Failure

Return:

- OCR processing error

Conversion Failure

Return:

- Conversion error with logs

Missing Tool

Return:

- Tool invocation failure
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7. Security Considerations

- Validate uploaded files
 - Restrict file size
 - Sanitize file paths
 - Prevent arbitrary command execution
 - Run conversion tools with least privilege
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8. Future Architecture

User ↓ Web UI ↓ Agent ↓ MCP Server ↓ OCR Tool Calibre Tool Metadata Tool Kindle Delivery Tool